

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Revision Test 2026

MATHEMATICS — SSC-I (9th Grade) — ANSWER KEY (Version 18 (B))

Syllabus: Unit 1 – Real Numbers, Unit 2 – Logarithms, Unit 4 – Factorization and Algebraic Manipulation, Unit 5 – Linear Equations and Inequalities, Unit 7 – Coordinate Geometry

Total Marks: 38

Time Allowed: 1 Hour 45 Minutes

Version 18 (B)

SECTION A — Answers to Multiple Choice Questions ($10 \times 1 = 10$ Marks)

Q1. $\sqrt{98} + \sqrt{8}$ is equal to:

Correct answer: **B) $9\sqrt{2}$** 1 mark

Simplify each radical first, then add like terms: $\sqrt{98} = \sqrt{(49 \times 2)} = 7\sqrt{2}$ and $\sqrt{8} = \sqrt{(4 \times 2)} = 2\sqrt{2}$, so the sum is $9\sqrt{2}$.

Adding the numbers under the radical signs ($98 + 8 = 106$) is **not** allowed: $\sqrt{a} + \sqrt{b} \neq \sqrt{a + b}$.

Source in her textbook: Unit 1, radicals (product rule $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$) — SLO: explain the concept of radicals and radicands; apply the laws of exponents. *Re-test of exam-17 Q2 ($\sqrt{48} + \sqrt{27}$ answered as 75).*

Q2. $\log 6 + \log 5$ is equal to:

Correct answer: **A) log 30** 1 mark

By the **product law**, $\log m + \log n = \log(mn)$. So $\log 6 + \log 5 = \log(6 \times 5) = \log 30$. Adding logs multiplies the numbers; it does not add them ($\log 11$ is wrong).

Source in her textbook: Unit 2, laws of logarithm (product law) — SLO: prove the four basic laws of logarithm and apply them.

Q3. $\log_3 (1/27)$ is equal to:

Correct answer: **C) -3** 1 mark

$\log_3 (1/27) = x$ means $3^x = 1/27 = 3^{-3}$, so $x = -3$. The value is negative because $1/27$ is less than 1.

Source in her textbook: Unit 2, logarithm of a number to the base a — SLO: define logarithm of a number to the base a .

Q4. The factors of $27m^3 + 64n^3$ are:

Correct answer: **D) $(3m + 4n)(9m^2 - 12mn + 16n^2)$** 1 mark

Sum of cubes: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ with $a = 3m$, $b = 4n$, giving $(3m + 4n)(9m^2 - 12mn + 16n^2)$. Rule of signs: the bracket sign is the same as in the question, the middle sign of the trinomial is the opposite.

Source in her textbook: Unit 4, factorization of $a^3 + b^3$ — SLO: factorize expressions of the type $a^3 \pm b^3$. *Re-test of exam-15 Q13 and exam-16 Q11 (sign pattern).*

Q5. The factors of $27 - 8y^3$ are:

Correct answer: **A) $(3 - 2y)(9 + 6y + 4y^2)$** 1 mark

Difference of cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ with $a = 3$, $b = 2y$, giving $(3 - 2y)(9 + 6y + 4y^2)$. Compare with Q4: same rule of signs.

Source in her textbook: Unit 4, factorization of $a^3 - b^3$ — SLO: factorize expressions of the type $a^3 \pm b^3$.

Q6. $|-9| - |-4|$ is equal to:

Correct answer: **B) 5** 1 mark

Absolute value is distance from zero, so it is never negative: $|-9| = 9$ and $|-4| = 4$. Then $9 - 4 = 5$. The bars are not brackets.

Source in her textbook: Unit 5, absolute value (" $|5| = 5$ and $|-5| = 5$ ") — SLO: define absolute value. *Re-test of exam-17 Q7 ($|-8| + |3|$ answered as -5).*

Q7. The solution of $-4 \leq 2x - 2 \leq 6$ is:

Correct answer: **C) $-1 \leq x \leq 4$** 1 mark

Add 2 to all three parts: $-2 \leq 2x \leq 8$. Divide all three parts by 2: $-1 \leq x \leq 4$. Keep the minus sign on the lower bound.

Source in her textbook: Unit 5, solving linear inequalities with rational coefficients — SLO: solve linear inequalities. *Re-test of exam-17 Q8 (minus sign dropped on the lower bound).*

Q8. If $x - 3 < 5$, then $x < 8$. The property of inequality used is the:

Correct answer: **A) additive property** 1 mark

The same number (3) is **added** to both sides without changing the direction of the inequality, so this is the **additive property**. Transitive is the chain rule ($a < b$ and $b < c \Rightarrow a < c$).

Source in her textbook: Unit 5, properties of inequalities — SLO: recognize properties of inequalities (trichotomy, transitive, additive and multiplicative). *Re-test of exam-16 Q18 (answered transitive).*

Q9. If the ordered pairs $(2a - 1, b + 3)$ and $(9, 5)$ are equal, then the value of a is:

Correct answer: **D) 5** 1 mark

Equal ordered pairs have equal corresponding elements: $2a - 1 = 9 \Rightarrow 2a = 10 \Rightarrow a = 5$ (and $b + 3 = 5 \Rightarrow b = 2$). 9 is the value of the whole element, not of a .

Source in her textbook: Unit 7, Miscellaneous Exercise 7 (equality of ordered pairs) — SLO: use conventions for coordinates in the Cartesian plane. *Re-test of exam-16 Q22 (answered with the element instead of a).*

Q10. The mid-point of the line segment joining $(-5, 4)$ and $(9, -8)$ is:

Correct answer: **B) (2, -2)** 1 mark

Mid-point = $((x_1 + x_2)/2, (y_1 + y_2)/2) = ((-5 + 9)/2, (4 + (-8))/2) = (4/2, -4/2) = (2, -2)$. (7, -6) is half the *difference*, and (4, -4) forgets to divide by 2.

Source in her textbook: Unit 7, mid-point formula — SLO: derive mid-point formula and calculate mid-point of a line segment. *Re-test of exam-17 Q10.*

Q	1	2	3	4	5	6	7	8	9	10
Key	B	A	C	D	A	B	C	A	D	B

SECTION B — Answers to Short Questions ($3 \times 4 = 12$ Marks)

Q.2(i)(a) Evaluate $\log 72$, naming the law at each step 3 marks

$$72 = 8 \times 9 = 2^3 \times 3^2 \quad 1$$

$$\log 72 = \log 2^3 + \log 3^2 \quad (\text{product law}) = 3 \log 2 + 2 \log 3 \quad (\text{power law}) \quad 1$$

$$= 3(0.3010) + 2(0.4771) = 0.9030 + 0.9542 = 1.8572 \quad 1$$

Marking note: the second mark needs both laws **named**. Correct working with no law named scores 2/3. Accept 72 written as $2 \times 2 \times 2 \times 3 \times 3$ with the product law named.

Q.2(i)(b) Characteristic of $\log 0.0567$ 1 mark

$0.0567 = 5.67 \times 10^{-2}$, so the characteristic is **-2**, written **2̄** ("bar 2"). Accept either form, as in her book's Example 9 ("characteristic = -2 or 2̄"). 1

Source in her textbook: Unit 2, laws of logarithm; characteristic and mantissa (Example 9) — SLOs: prove and apply the four basic laws; define characteristic and mantissa.

Q.2(ii)(a) Solve $|x - 3| + 4 = 9$ and verify both solutions 2 marks

$$|x - 3| = 5 \Rightarrow x - 3 = 5 \text{ or } x - 3 = -5 \Rightarrow x = 8 \text{ or } x = -2 \quad 1$$

$$\text{Check: } |8 - 3| + 4 = 5 + 4 = 9 \checkmark \quad |-2 - 3| + 4 = |-5| + 4 = 9 \checkmark \quad 1$$

Marking note: the second mark is for the verification. Both roots with no check scores 1/2.

Q.2(ii)(b) Solve $3(x - 1) \geq 2x + 4$, stating the property used 2 marks

$$3x - 3 \geq 2x + 4 \quad (\text{distributive law})$$

$$3x - 3 - 2x \geq 2x + 4 - 2x \Rightarrow x - 3 \geq 4 \quad (\text{additive property})$$

$$x - 3 + 3 \geq 4 + 3 \Rightarrow x \geq 7 \quad (\text{additive property}) \quad 1$$

Property named correctly 1. Correct answer with no property named scores 1/2.

Source in her textbook: Unit 5, absolute value equations; properties of inequalities — SLOs: solve the equation involving absolute value; recognize properties of inequalities; solve linear inequalities.

